

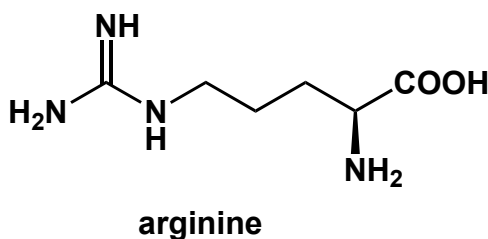
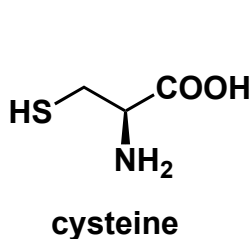
1. To 100 mL of 1 M acetic acid solution is added 0.05 mol NaOH solid, which results in Solution A. Calculate the pH of Solution A. If 0.01 mol NaOH solid is further added to Solution A, calculate the pH of the solution. What is the pH if 0.01 mol H₂SO₄ is added to Solution A? Assuming the volume of the solution does not change during these processes. Acetic acid has a pK_a of 4.76.

2. The pK_a values of the following two amino acids are as follows:

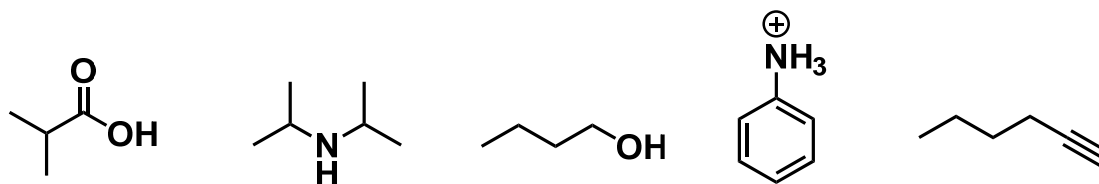
(a) cysteine: 1.8, 8.3, and 10.8

(b) arginine: 2.2, 9.0, and 13.2

Assign these pK_a values to the functional group in each amino acid and draw the most abundant structure that each molecule will have at pH 1, 7, 10, and 14.



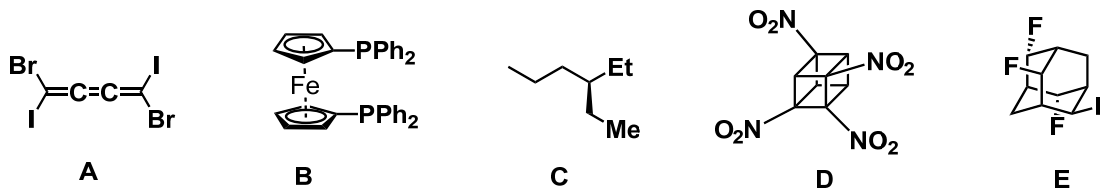
3. Which of these bases should be chosen to deprotonate the following molecules? Very strong bases are more challenging to handle so you should aim to use a base which is just strong enough for the job, but not unnecessarily strong.



Choice of bases:

KOH, NaH, butyl lithium (BuLi), NaHCO₃.

4. Which of the following molecules are not chiral? Explain your answer.



5. Draw the most stable conformation for the following molecules.



6. A reaction generates four products: A and its enantiomer (C), and B and its

enantiomer (D). A and B are diastereomers. The total yield for all products is 78%. The diastereomeric ratio (*d.r.*) of A to B is 2:1. The ee% for A is 90% and the ee% for B is 80%. Calculate the individual yield for each product.